

## Section 1 - Identification

**Product Name** Cetrimide Agar (MLT)

|                                |                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>R110100, R110292</b>                                                                              |
| <b>Address</b>                 | ThermoFisher Scientific Australia Pty Ltd<br>5 Caribbean Drive, Scoresby<br>VICTORIA 3179, Australia |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>03 9757 4559 or +613 9757 4559</b>                                            |
| <b>Telephone / Fax Numbers</b> | Tel: 1300 735 292<br>Fax: 1800 067 639                                                               |
| <b>E-mail address</b>          | ANZinfo@thermofisher.com                                                                             |

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as not hazardous according to criteria of Safe Work Australia.

**Physical hazards**  
No hazards identified

**Health hazards**  
No hazards identified

**Environmental hazards**  
No hazards identified

### Label Elements

### **Other information**

No information available  
This product does not contain any known or suspected endocrine disruptors

## Section 3 - Composition and Information on Ingredients

| Component           | CAS No     | Weight % |
|---------------------|------------|----------|
| Tetradonium bromide | 1119-97-7  | 0.44     |
| Peptones, casein    | 91079-40-2 | 46.4     |
| Potassium sulfate   | 7778-80-5  | 22.09    |
| Magnesium sulfate   | 7487-88-9  | 3.09     |
| Yeast, ext.         | 8013-01-2  | 2.2      |
| Sodium carbonate    | 497-19-8   | 0.33     |
| Agar                | 9002-18-0  | 25.41    |
| Glycerin            | 56-81-5    | 0.94     |

## Section 4 - First Aid Measures

|                                            |                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | Remove to fresh air.                                                                                                                             |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| <b>Skin Contact</b>                        | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.                                            |
| <b>Eye Contact</b>                         | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician.                             |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                             |
| <b>Most important symptoms and effects</b> | No information available.                                                                                                                        |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                           |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

### Extinguishing media which must not be used for safety reasons

No information available.

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Emergency procedures

Ensure adequate ventilation.

### Environmental Precautions

See Section 12 for additional Ecological Information.

### Methods for Containment and Clean Up

**Clean-up methods - small spillage**

Soak up with inert absorbent material.

**Clean-up methods - large spillage**

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Ensure adequate ventilation.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep container tightly closed in a dry and well-ventilated place.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

**DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component | Australia                 | New Zealand WEL           | ACGIH TLV | The United Kingdom                         | Germany                                                                                                                                            |
|-----------|---------------------------|---------------------------|-----------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Glycerin  | TWA: 10 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup> |           | TWA: 10 mg/m <sup>3</sup> 8 hr (mist only) | TWA: 200 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 200 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 400 mg/m <sup>3</sup> |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material    | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|-------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Disposable gloves | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices (or AUS/NZ equivalent)

When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

No information available.

## Section 9 - Physical and Chemical Properties

**Information on basic physical and chemical properties****Appearance****Physical State**

Low melting solid

**Odor**

No information available

**Odor Threshold**

No data available

**pH**

No information available

**Melting Point/Range**

No data available

**Softening Point**

No data available

**Boiling Point/Range**

No information available

**Flash Point**

No information available

**Method -** No information available**Evaporation Rate**

No data available

**Flammability (solid,gas)**

No information available

**Explosion Limits**

No data available

**Vapor Pressure**

No data available

**Vapor Density**

No data available

(Air = 1.0)

**Specific Gravity / Density**

No data available

**Bulk Density**

No data available

**Water Solubility**

No information available

**Solubility in other solvents**

No information available

**Partition Coefficient (n-octanol/water)****Component****log Pow**

Tetradonium bromide

2.2

Peptones, casein

0.3

Glycerin

-1.75

**Autoignition Temperature**

No data available

**Decomposition Temperature**

No data available

**Viscosity**

No data available

**Explosive Properties**

No information available

**Oxidizing Properties**

No information available

**Other information**

## Section 10 - Stability and Reactivity

**Reactivity** None known, based on information available

**Stability** Stable under normal conditions.

**Conditions to Avoid** Heat, flames and sparks.

**Incompatible Materials** None known.

**Hazardous Decomposition Products** None under normal use conditions.

**Hazardous Polymerization** No information available.

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

**(a) acute toxicity;**

**Oral** No data available  
**Dermal** No data available  
**Inhalation** No data available

| Component           | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation              |
|---------------------|---------------------------|---------------------------|------------------------------|
| Tetradonium bromide | LD50 = 1000 mg/kg ( Rat ) |                           |                              |
| Potassium sulfate   | LD50 = 6600 mg/kg ( Rat ) | LD50 > 2000 mg/kg ( Rat ) |                              |
| Sodium carbonate    | 2800 mg/kg ( Rat )        | > 2000 mg/kg (rabbit)     | 2.3 mg/l 2h (Rat)            |
| Agar                | LD50 = 11 g/kg ( Rat )    |                           |                              |
| Glycerin            | 12600 mg/kg ( Rat )       | > 10 g/kg ( Rabbit )      | > 2.75 mg/L/4h ( Rat )(mist) |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

**(d) respiratory or skin sensitization;**

**Respiratory** No data available  
**Skin** No data available

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;** No data available

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** No data available

**(h) STOT-single exposure;** No data available

**(i) STOT-repeated exposure;** No data available

## Target Organs

No information available.

## (j) aspiration hazard;

No data available

Symptoms / effects, both acute and delayed

No information available

## Section 12 - Ecological Information

## Ecotoxicity effects

Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component           | Freshwater Fish                                                                                                                                              | Water Flea                                           | Freshwater Algae                                  | Microtox                                                          |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------|
| Tetradonium bromide | LC50 (Danio rerio (zebra fish)): 1,81 mg/l (96 h)                                                                                                            |                                                      | LC50 (Danio rerio (zebra fish)): 1,81 mg/l (96 h) | LC50 (Danio rerio (zebra fish)): 1,81 mg/l<br>Exposure time: 96 h |
| Potassium sulfate   | LC50: 510 - 880 mg/L, 96h static (Pimephales promelas)<br>LC50: = 653 mg/L, 96h (Lepomis macrochirus)<br>LC50: = 3550 mg/L, 96h static (Lepomis macrochirus) | EC50: 890 mg/L 48h                                   | EC50: 2900 mg/L 72h                               |                                                                   |
| Magnesium sulfate   | LC50: 2610 - 3080 mg/L, 96h static (Pimephales promelas)                                                                                                     | EC50: 266.4 - 417.3 mg/L, 48h Static (Daphnia magna) | EC50: = 2700 mg/L, 72h (Desmodesmus subspicatus)  | = 84000 mg/L EC50 Photobacterium phosphoreum 30 min               |
| Sodium carbonate    | Lepomis macrochirus: LC50: 300 mg/L/96h<br>Gambusia affinis: LC50: 740 mg/L/96h                                                                              | EC50: = 265 mg/L, 48h (Daphnia magna)                |                                                   | -                                                                 |
| Glycerin            | LC50: 51 - 57 mL/L, 96h static (Oncorhynchus mykiss)                                                                                                         |                                                      |                                                   |                                                                   |

## Persistence and Degradability

No information available

## Bioaccumulative Potential

No information available

| Component           | log Pow | Bioconcentration factor (BCF) |
|---------------------|---------|-------------------------------|
| Tetradonium bromide | 2.2     | No data available             |
| Peptones, casein    | 0.3     | No data available             |
| Glycerin            | -1.75   | No data available             |

## Mobility

No information available.

## Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## Persistent Organic Pollutant

This product does not contain any known or suspected substance

## Ozone Depletion Potential

This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

## Waste from Residues/Unused Products

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

## Contaminated Packaging

Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.

## Other Information

Chemical wastes should be disposed through a licensed commercial waste collection service.

## Section 14 - Transport Information

|                               |                                 |
|-------------------------------|---------------------------------|
| <u>IMDG/IMO</u>               | Not regulated                   |
| <u>ADG</u>                    | Not regulated                   |
| <u>IATA</u>                   | Not regulated                   |
| <b>Environmental hazards</b>  | No hazards identified           |
| <b>Special Precautions</b>    | No special precautions required |
| <b>Additional information</b> | None known                      |

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations                      **Australia**

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                     | Standard for the Uniform Scheduling of Medicines and Poisons                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Magnesium sulfate - 7487-88-9 | Schedule 3 listed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Sodium carbonate - 497-19-8   | Schedule 5 listed - being the carbonate, silicate or phosphate salts of sodium or potassium alone or in any combination: in solid orthodontic device cleaning preparations, the pH of which as an in-use aqueous solution is >11.5, in solid automatic dishwashing preparations, the pH of which in a 500 g/L aqueous solution or mixture is >11.5 but ≤12.5; in other solid preparations, the pH of which in a 10 g/L aqueous solution is >11.5, or in liquid or semi-solid preparations, the pH of which is >11.5, unless: in food additive preparations for domestic use, or in automatic dish washing preparations for domestic use with a pH >12.5; except when separately specified in these Schedules<br>Schedule 5 listed - being the Carbonate, Silicate or Phosphate salts of Sodium or Potassium alone or in any combination: in solid orthodontic device cleaning preparations as an in-use aqueous solution; in solid automatic dishwashing preparations in a 500 g/L aqueous solution or mixture but with pH ≤12.5; in other solid preparations in a 10 g/L aqueous solution, or in liquid or semi-solid preparations, unless in food additive preparations for domestic use, or in automatic dish washing preparations for domestic use with a pH >12.5; except when separately specified in these Schedules<br>Schedule 6 listed - being the carbonate, silicate or phosphate salts of sodium or potassium alone or in any combination for non-domestic use: in solid automatic dishwashing preparations, the pH of which in a 500 g/L aqueous solution or mixture is >12.5, or in liquid or semi-solid automatic dishwashing preparations, the pH of which is >12.5<br>Schedule 10 listed |

#### **Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                       | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|---------------------------------|-------------------------------------------------------------|------------------------|
| Tetradonium bromide - 1119-97-7 | Present                                                     | -                      |
| Potassium sulfate - 7778-80-5   | Present                                                     | -                      |
| Magnesium sulfate - 7487-88-9   | Present                                                     | -                      |
| Yeast, ext. - 8013-01-2         | Present                                                     | -                      |
| Sodium carbonate - 497-19-8     | Present                                                     | -                      |

|                    |         |   |
|--------------------|---------|---|
| Agar - 9002-18-0   | Present | - |
| Glycerin - 56-81-5 | Present | - |

**Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

**Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component           | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCs | ISHL | IECSC | KECL       |
|---------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|------------|
| Tetradonium bromide | X    | X     | 214-291-9 | -      | X    | X   | -    | -     | X    | X    | X     | KE-34687   |
| Peptones, casein    | -    | X     | 293-428-4 | -      | -    | -   | -    | X     | -    |      | X     | -          |
| Potassium sulfate   | X    | X     | 231-915-5 | -      | X    | X   | -    | X     | X    | X    | X     | KE-29200   |
| Magnesium sulfate   | X    | X     | 231-298-2 | -      | X    | X   | -    | X     | X    | X    | X     | KE-22752   |
| Yeast, ext.         | X    | X     | 232-387-9 | -      | X    | X   | -    | X     | -    |      | X     | KE-05-1355 |
| Sodium carbonate    | X    | X     | 207-838-8 | -      | X    | X   | -    | X     | X    | X    | X     | KE-31380   |
| Agar                | X    | X     | 232-658-1 | -      | X    | X   | -    | X     | -    |      | X     | KE-00275   |
| Glycerin            | X    | X     | 200-289-5 | -      | X    | X   | -    | X     | X    | X    | X     | KE-29297   |

**Legend:** X - Listed. '-' - Not Listed. XU - Indicates a substance exempt from reporting under the Inventory Update Rule, i.e. Partial Updating of the TSCA Inventory Data Base Production and Site Reports (40 CFR 710(B)). **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Not applicable.

| Component           | CAS No     | OECD HPV       | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|---------------------|------------|----------------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Tetradonium bromide | 1119-97-7  | Not applicable | Not applicable                             | E1                                                                                        | Not applicable                                                                           |
| Peptones, casein    | 91079-40-2 | Not applicable | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |
| Potassium sulfate   | 7778-80-5  | Listed         | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |



|                   |           |                |                |                |                |
|-------------------|-----------|----------------|----------------|----------------|----------------|
| Magnesium sulfate | 7487-88-9 | Listed         | Not applicable | Not applicable | Not applicable |
| Yeast, ext.       | 8013-01-2 | Not applicable | Not applicable | Not applicable | Not applicable |
| Sodium carbonate  | 497-19-8  | Listed         | Not applicable | Not applicable | Not applicable |
| Agar              | 9002-18-0 | Not applicable | Not applicable | Not applicable | Not applicable |
| Glycerin          | 56-81-5   | Listed         | Not applicable | Not applicable | Not applicable |

## Authorisation/Restrictions according to EU REACH

| Component        | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Sodium carbonate | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

## Section 16 - Other Information

Legend

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>AICS</b> - Australian Inventory of Chemical Substances                                            | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                                            |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists Predicted No Effect Concentration (PNEC)            |
| <b>IARC</b> - International Agency for Research on Cancer                                            | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>ADG</b> - Australian Code for the Transport of Dangerous Goods by Road and Rail                                           |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>NZS 5433:2020</b> - Transport of Dangerous Goods on Land                                          | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  |                                                                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

## Revision Date

05-Jul-2023

## Revision Summary

Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**